Research the Top 10 LLM's and their differences, builds, features, strengths, weaknesses and most used application.

I'll now analyze the collected research and create a comprehensive report on the top 10 LLMs with their differences, features, strengths, weaknesses, and applications.

Top 10 Large Language Models October 2025: Comprehensive Analysis

Date: Friday, October 24, 2025, 8:25 AM AWST

TL;DR

- **GPT-5** (OpenAI) leads overall with unified chat+thinking modes, 94.6% AIME 2025, 74.9% SWE-bench, and state-of-the-art multimodal capabilities[1] [2] [3]
- **Claude 4 Opus** (Anthropic) dominates sustained coding with 74.5% SWE-bench, excels in long-running agentic tasks (30+ hours continuous), and leads in safety alignment[4] [5] [6]
- **Gemini 2.5 Pro** (Google DeepMind) offers unprecedented 2M token context (production-ready), strongest multimodal integration, and leads on web dev (WebDev Arena +147 Elo)[7] [8] [9]
- **DeepSeek-V3.1/V3.2-Exp** (DeepSeek) provides best cost-efficiency ($0.07/M tokens cached), open-source leader with 671B params (37B active MoE), hybrid thinking modes[10] [11] [12]
- **Qwen3-235B-A22B** (Alibaba) excels in multilingual (119 languages), thinking+non-thinking hybrid modes, strongest open agent capabilities, and MoE efficiency[13] [14] [15]
- **Kimi K2** (Moonshot AI) introduces 1T params (32B active), leads LiveCodeBench v6 at 53.7%, ultra-low API cost ($0.15/M input), trained with novel MuonClip optimizer[16] [17] [18]
- **Llama 4 Maverick/Scout** (Meta) offers first Llama MoE architecture, native multimodality, 10M token context (Scout), open-weight under 700M MAU limit[19] [20] [21]
- **Grok 4** (xAI) achieves 50.7% on Humanity's Last Exam (Heavy mode), 100% HMMT 2025, 66.6% ARC-AGI, best business simulation performance[22] [23] [24]
- **Claude Sonnet 4.5** (Anthropic) leads computer use (61.4% OSWorld), maintains focus 30+ hours, state-of-the-art coding efficiency, optimal capability-cost balance[25] [5]
- **Mistral Medium 3** (Mistral AI) delivers 90% of Claude Sonnet 3.7 performance at 8× lower cost, deployable on 4 GPUs, strong enterprise focus[26] [27] [28]

A) CHANGES FOUND — IMPACTFUL FIRST

Major Architecture Shifts (2025)

1. Hybrid Reasoning Systems (NEW)

- GPT-5, Qwen3, and DeepSeek-V3.1 now offer **seamless switching** between "thinking" (extended reasoning) and "non-thinking" (fast) modes within single models[29] [13] [10]
- Represents fundamental shift from separate model families to unified adaptive intelligence[30] [3]

2. Mixture-of-Experts (MoE) Dominance

- Llama 4 (first Llama MoE), Kimi K2 (1T params), Qwen3, DeepSeek, and Grok 4 all use MoE for efficiency[31] [17] [19]
- Enables massive parameter counts (1T+) with only 3-5% active per inference, reducing cost 50-80%[11] [16]

3. Extended Context Windows

- **Gemini 2.5 Pro**: 2M tokens production-ready (testing 6 hours video)[8] [32] [7]
- **Llama 4 Scout**: 10M tokens context[20] [21] [19]
- **Qwen3-Next**: up to 1M tokens extensible[33] [34]

4. Native Multimodality Maturation

- All frontier models (GPT-5, Claude 4, Gemini 2.5, Llama 4) now **natively process text, images, video, audio** from training[35] [1] [7] [19]
- Early fusion architecture replaces separate encoders for superior cross-modal reasoning[36] [37]

5. Agentic Optimization

- Kimi K2, Qwen3, and DeepSeek-V3.1 specifically **trained for tool use, function calling, and autonomous workflows**[17] [33] [16]
- New benchmarks: τ2-bench telecom (96.7% GPT-5), SWE-bench Verified (74.9% GPT-5, 74.5% Opus 4.1)[2] [1] [4]

Pricing Disruption (2025)

DeepSeek API price cut 50%+: Now $0.07/M input (cache hit), $0.28 (cache miss), $0.42 output[12] [11]

- **Kimi K2**: $0.15/M input, $2.50/M output—undercutting OpenAI/Anthropic massively[38] [17]
- **Mistral Medium 3**: 8× cheaper than comparable models while matching 90% of Sonnet 3.7[26]

Chinese Open-Source Surge

DeepSeek, Qwen3 (Alibaba), and Kimi K2 (Moonshot AI) now **competitive with or exceeding GPT-4.1** on coding/math benchmarks while remaining open-weight[39] [40] [41] [16]

- Represents strategic shift in AI power balance toward open development

B) CAPABILITY GRID

Model	Provider	Context	Params Active	Best For	Top Benchmark	API Cost (Input/Output per 1M)	Open?
GPT-5	OpenAI	400K	Unknown	Overall intelligence, math, multimodal	94.6% AIME 2025 [1]	$1.25/$10 [42]	No
Claude 4 Opus	Anthropic	200K	Unknown	Long coding sessions, sustained agents	74.5% SWE-bench [4]	$15/$75	No
Gemini 2.5 Pro	Google	2M	Unknown	Long-context, multimodal, web dev	84.8% VideoMME [7]	Varies	No
DeepSeek-V3.2	DeepSeek	128K	37B (671B total)	Cost-efficiency, reasoning	88.4% AIME [10]	$0.07/$0.42 [12]	Yes
Qwen3-235B	Alibaba	262K	22B (235B total)	Multilingual, agents, reasoning	83.0% MMLU-Pro [33]	$0.27/$1.10 [43]	Yes
Kimi K2	Moonshot AI	128K	32B (1T total)	Agentic coding, tool use	53.7% LiveCodeBench [16]	$0.15/$2.50 [17]	Yes
Llama 4 Maverick	Meta	1M	17B (400B total)	Multimodal, open deployment	200 languages [19]	Self-host	Yes*
Grok 4	xAI	256K	Unknown	Academic reasoning, business sim	50.7% HLE [22]	$3/$15 [23]	No
Claude Sonnet 4.5	Anthropic	200K	Unknown	Balanced coding, computer use	61.4% OSWorld [25]	Lower than Opus	No
Mistral Medium 3	Mistral AI	128K	Unknown	Cost-efficient enterprise	90% of Sonnet 3.7 [26]	$0.4/$2 [26]	No

*Llama 4 requires separate license above 700M MAU [21] [44]

C) DECISION MATRIX — BY USE CASE

 Best Overall Single Choice (October 2025)

→ **GPT-5** — Unified system with automatic thinking mode selection, state-of-the-art across math (94.6% AIME), coding (74.9% SWE-bench), multimodal (84.2% MMMU), and health (46.2% HealthBench Hard)[3] [1] [2]

- **Why:** Automatic routing between fast/deep reasoning eliminates model-switching complexity[29] [30]
- **Limitation:** Highest cost at $1.25/$10 per M tokens[42]

 Agentic Coding & Long-Horizon Development

→ **Claude 4 Opus** — Sustained 30+ hours continuous performance, 74.5% SWE-bench Verified, excels at multi-file refactoring and precise debugging[5] [45] [4]

- **Runner-up:** GPT-5 (74.9% SWE-bench, better short tasks)[2]
- **Budget option:** Kimi K2 (53.7% LiveCodeBench v6, $0.15/$2.50)[18] [16]

 Long-Context & Document Analysis

→ **Gemini 2.5 Pro** — Production-ready 2M tokens, native multimodality, processes 6+ hours video[9] [7] [8]

- **Why:** Only model with truly production-scale long context (Llama Scout's 10M still experimental)[19]
- **Use cases:** Full codebase review, legal document analysis, video understanding[32] [46]

 Multilingual & Global Deployment

→ **Qwen3-235B-A22B** — 119 languages, strongest multilingual instruction-following, excellent tool calling[14] [15] [13]

- **Runner-up:** Gemini 2.5 Pro (140+ languages, but less open)[8]
- **Agent focus:** Qwen3 leads BFCL-v3 function calling among open models[33]

 Cost-Efficient / Budget-Conscious

→ **DeepSeek-V3.2-Exp** — $0.07/M (cache hit), 50%+ price cut, maintains V3.1 performance with sparse attention[11] [12]

- **Runner-up:** Kimi K2 ($0.15/$2.50, better agentic coding)[17]
- **Enterprise:** Mistral Medium 3 (8× cheaper, 4-GPU deployment)[26]

 Best Open-Weight (Self-Hosting)

→ **Qwen3-235B-A22B (Thinking)** — Strongest reasoning + agent capabilities, 119 languages, MoE efficiency[15] [13] [33]

- **Runner-up:** DeepSeek-V3.1 (671B params, hybrid modes, strong math)[10] [11]
- **Smallest powerful:** Llama 4 Scout (17B active, multimodal, 10M context)[20] [19]

Multimodal (Text + Vision + Video)

→ **Gemini 2.5 Pro** — Best video understanding (84.8% VideoMME), spatial reasoning, native audio-visual[47] [7] [9]

- **Runner-up:** GPT-5 (84.2% MMMU, better text reasoning)[1] [3]
- **Open:** Llama 4 Maverick (native early fusion, 400B params)[37] [19]

Research & PhD-Level Reasoning

→ **Grok 4 Heavy** — 50.7% Humanity's Last Exam (multi-agent), 100% HMMT 2025, 66.6% ARC-AGI[24] [48] [22]

- **Runner-up:** GPT-5 Pro (extended thinking, 26.3% FrontierMath)[1] [2]
- **Note:** Both excel at test-time compute scaling

Enterprise Deployment (On-Prem)

→ **Mistral Medium 3** — 4-GPU minimum, VPC/on-prem support, custom post-training, 90% of Sonnet 3.7 performance[28] [26]

- **Runner-up:** Qwen3 (fully open, strong enterprise features)[15] [33]
- **Large-scale:** Llama 4 (Meta support, proven infrastructure)[44] [21]

Computer Use / UI Automation

→ **Claude Sonnet 4.5** — 61.4% OSWorld (up from 42.2% four months prior), browser navigation, form filling[25] [5]

- **Why:** Massive improvement in real-world computer task execution
- **Use case:** RPA, browser automation, spreadsheet manipulation

D) WEAVED SOLUTION BLUEPRINT

Example: Enterprise Code Assistant with Multi-Model Routing

Architecture

```
User Query → Router (GPT-5 mini) → Task Classification
    ↓
├─ Quick Answers (< 100 tokens) → Gemini 2.5 Flash (latency-optimized)
├─ Standard Coding → Claude Sonnet 4.5 (balanced cost/performance)
├─ Long Sessions (> 1hr) → Claude 4 Opus (sustained focus)
├─ Multi-file Refactor → GPT-5 (parallel tool calls, broader context)
├─ Cost-Sensitive Batch → DeepSeek-V3.2 (50% cheaper, good quality)
└─ Research/Math → Grok 4 Heavy (extended reasoning)
```

Routing Logic

1. **Fast Track** (< 5 sec): Gemini 2.5 Flash for definitions, syntax checks, simple fixes
2. **Standard** (5-60 sec): Claude Sonnet 4.5 for code generation, debugging, unit tests
3. **Deep Thinking** (> 60 sec): GPT-5 thinking mode or Grok 4 for architecture decisions
4. **Long Context**: Gemini 2.5 Pro if input > 128K tokens (full repo analysis)
5. **Budget Mode**: DeepSeek-V3.2 for internal tools, high-volume batch jobs

Retrieval Layer

- **Embedding**: OpenAI `text-embedding-3-large` or Mistral Embed
- **Vector DB**: Pinecone/Weaviate with 8K chunk size
- **Reranking**: Claude Haiku 3 (fast, cost-efficient)

Evaluator Model

- **GPT-5 mini** as judge for code quality, safety checks (Scale MultiChallenge: 69.6%)[2]
- **Fallback**: If confidence < 0.7, escalate to Opus 4 for review

Fallbacks

1. **Primary fails** → Retry with alternate (Sonnet → GPT-5)
2. **Rate limit** → Queue to DeepSeek or Qwen3
3. **Context overflow** → Auto-switch to Gemini 2.5 Pro (2M context)

E) BENCH/COST PLAN

Test Prompts (5-8 Representative Tasks)

1. **Coding (SWE-bench style)**: "Fix memory leak in React useEffect with WebSocket connection cleanup"
2. **Math Reasoning**: "Solve AIME 2025 #15: Complex number sequence convergence proof"
3. **Long Context**: "Summarize architectural decisions across 50-file Python codebase (300K tokens)"
4. **Multimodal**: "Analyze this architecture diagram + code screenshot, identify security vulnerabilities"
5. **Tool Use**: "Book flight to Tokyo, check calendar conflicts, send meeting invite, update expense report"
6. **Multilingual**: "Translate technical documentation Spanish → Japanese, maintain code formatting"
7. **Agent Task**: "Research competitor pricing, generate comparison table, draft email to sales team"
8. **Computer Use**: "Navigate to HubSpot, extract last 10 leads, create CSV, upload to Sheets"

Success Metrics

Metric	Target	Measurement
Accuracy	> 85% correct	Human expert evaluation (0-100 scale)
Latency	< 30 sec (standard), < 120 sec (thinking)	Time to first/complete token
Cost per Task	< $0.05 (standard), < $0.50 (complex)	Input + output tokens × API rate
Context Efficiency	< 50% token waste	Relevant tokens / total tokens used
Tool Call Accuracy	> 90% correct args	Structured output validation

A/B Harness Notes

Framework: LangSmith or PromptLayer for logging

- **Blind evaluation**: 2-3 expert reviewers rate outputs without knowing model
- **Statistical significance**: Minimum 100 runs per model/prompt pair
- **Cost tracking**: Real-time budget alerts at $10/$50/$100 thresholds
- **Latency percentiles**: Track P50, P95, P99 for production planning

Run weekly to catch model updates (e.g., GPT-5 → GPT-5.1 drift)

F) INTEGRATION NOTES

Exact API Model Slugs

Provider	Model Slug	Endpoint	Rate Limits	Notes
OpenAI	`gpt-5`, `gpt-5-thinking`, `gpt-5-pro`	`https://api.openai.com/v1/chat/completions`	10K TPM (Tier 1)	Supports `verbosity` param (low/med/high) [2]
Anthropic	`claude-opus-4-20250804`, `claude-sonnet-4-5-20250928`	`https://api.anthropic.com/v1/messages`	40K TPM (base)	Extended thinking with tool use [4] [25]
Google	`gemini-2.5-pro`, `gemini-2.5-flash`	`https://generativelanguage.googleapis.com/v1beta/`	Varies by tier	Thinking budget control [8] [9]
DeepSeek	`deepseek-v3-2-exp`	`https://api.deepseek.com/v1/chat/completions`	60 RPM	OpenAI-compatible [11] [12]

Provider	Model Slug	Endpoint	Rate Limits	Notes
Alibaba	`qwen3-235b-a22b-thinking-2507`	Via Alibaba Cloud or HuggingFace	N/A (self-host)	Hybrid thinking/non-thinking [13] [33]
Moonshot	`kimi-k2-instruct`	`https://api.moonshot.cn/v1/chat/completions`	Custom tier	MCP server support [17] [18]
xAI	`grok-4-0709`, `grok-4-heavy`	`https://api.x.ai/v1/chat/completions`	60 RPM, 16K TPM	Heavy mode requires special access [22] [23]
Mistral	`mistral-medium-3`	`https://api.mistral.ai/v1/chat/completions`	Varies	VPC deployment option [26] [28]

Provider-Specific Quirks

OpenAI (GPT-5)

- **Verbosity control**: New `verbosity: "low"/"medium"/"high"` parameter for response length [49] [2]
- **Thinking toggle**: Model auto-decides, or force with system prompt "Use extended reasoning" [30] [3]
- **Parallel tools**: Supports 40+ simultaneous tool calls in single turn [2]

Anthropic (Claude 4)

- **Extended thinking**: Enable with `thinking_budget: "high"` in API call [4] [5]
- **Memory files**: Opus 4 autonomously creates markdown memory for long sessions [5]
- **Thinking summaries**: Auto-condenses long reasoning chains (5% of cases) [5]

Google (Gemini 2.5)

- **Context length**: Specify `max_tokens: 2000000` for full 2M window [7] [8]
- **Thinking mode**: `generation_config.thinking_duration` (low/medium/high) [50] [9]
- **Grounding**: Enable with `tools: [{"google_search": {}}]` [8]

DeepSeek

- **Sparse attention**: V3.2-Exp uses DSA automatically, no config needed [12] [11]
- **Hybrid modes**: Specify `<think>` tags to force thinking mode [10] [11]
- **Cache optimization**: First 1K tokens cached free, then $0.07/M [12]

Qwen3

- **Mode switching**: No explicit toggle; model auto-detects if thinking needed [13] [14]
- **MCP integration**: Native Model Context Protocol support for tool orchestration [51] [15]
- **Context extension**: YaRN RoPE scaling extends to 1M+ tokens [52] [51]

Kimi K2

- **Agentic tasks**: Optimized for multi-step workflows, may over-plan simple queries[16] [38]
- **Tool calling**: Requires explicit function schemas in prompt[18] [17]
- **Latency**: Slower than advertised for 1T param model, use streaming[18]

Grok 4

- **Heavy mode**: Separate endpoint, multi-agent system, 3-5× slower[23] [22]
- **Cost**: Cache hits reduce to $0.75/M input (vs $3.00 standard)[23]
- **Context**: Extended pricing kicks in above 128K tokens[23]

Mistral

- **Deployment**: On-prem requires minimum 4× H100 GPUs[28] [26]
- **Custom training**: Contact for post-training on proprietary data[26]
- **JSON mode**: Native structured output with `response_format: {"type": "json_object"}`[28]

Function Calling / Structured Outputs

Best for tools: Qwen3 (BFCL-v3: 70.3%), GPT-5 (τ2-bench: 96.7%), Claude 4 (native execution)[33] [2]

- **Format**: OpenAI-compatible function schemas work across most APIs
- **Parallel execution**: GPT-5, Claude 4 support simultaneous multi-tool calls[5] [2]

G) CHANGELOG — DATED LLM RADAR LOG

October 2025

2025-10-13: **Anthropic** expanded partnership with Salesforce for enterprise Claude deployment in regulated industries[53]

2025-10-08: **Reflection AI** raised $2B, built frontier LLM training stack, team from PaLM/Gemini/AlphaGo[54]

2025-10-06: **BigCode Arena** launch shows `o3-mini`, `o1-mini` lead coding; Qwen2.5, Llama-3.3-70B lag proprietary models[55]

September 2025

2025-09-29: **DeepSeek-V3.2-Exp** released with sparse attention (DSA), **50%+ API price cut** to $0.07/M (cache hit)[11] [12]

2025-09-28: **Claude Sonnet 4.5** announced as "best coding model," 61.4% OSWorld, 30+ hour sustained focus[25]

2025-09-24: **OpenAI GDPval** benchmark shows GPT-5 approaching expert-level work quality (100× faster, 100× cheaper)[56]

2025-09-21: **Qwen3-Next-80B-A3B** released with hybrid attention (Gated DeltaNet + Attention), extreme MoE sparsity[34] [33]

August 2025

2025-08-29: **DeepSeek-V3.1** launched with hybrid thinking/non-thinking modes, 128K context, improved tool calling[10]

2025-08-26: **Anthropic-OpenAI safety pilot** shows Claude Opus 4/Sonnet 4 match o3 on system prompt protection[57]

2025-08-14: **Epoch AI** reports frontier AI performance accessible on consumer GPUs (RTX 5090) within 6-12 months[58]

2025-08-07: **OpenAI GPT-5** official release — unified chat/thinking system, 94.6% AIME, 74.9% SWE-bench[3] [1] [2]

2025-08-06: **GPT-5 System Card** published detailing safety improvements, 26% lower hallucination rate vs GPT-4o[59] [60]

2025-08-05: **Qwen3** (Alibaba) released with 235B params (22B active), thinking/non-thinking hybrid, 119 languages[14] [13] [15]

2025-08-04: **Claude Opus 4.1** upgrade announced, 74.5% SWE-bench Verified, improved agentic research[4]

July 2025

2025-07-23: **Kimi K2** (Moonshot AI) released as 1T param MoE, 53.7% LiveCodeBench v6, $0.15/$2.50 API pricing[16] [17]

2025-07-09: **Grok 4** (xAI) launched with 50.7% Humanity's Last Exam (Heavy mode), 256K context, $3/$15 API[48] [22] [23]

June 2025

2025-06-16: **Gemini 2.5 Pro/Flash** general availability, 2.5 Flash-Lite preview as most cost-efficient 2.5 model[9]

May 2025

2025-05-28: **DeepSeek-R1-0528** released with additional reasoning optimizations over original R1[11]

2025-05-22: **Claude 4 Opus** launched as "best coding model in the world," 72.5% SWE-bench, 43.2% Terminal-bench[6] [5]

2025-05-07: **Mistral Medium 3** announced at 8× lower cost, 90% of Sonnet 3.7 performance, 4-GPU deployment[26]

April 2025

2025-04-28: **Qwen3** initial release with thinking mode, MoE architecture, 100+ languages[51] [13] [14]

2025-04-20: **Llama 4 Maverick & Scout** released — first Llama MoE, native multimodality, 10M context (Scout)[37] [19] [20]

March 2025

2025-03-25: **Gemini 2.5 Pro** official launch with 1M token context, enhanced reasoning, 63.8% SWE-bench[50] [7] [8]

2025-03-24: **Gemini 2.5** announced as "most intelligent AI model," native multimodality, thinking capabilities[50]

2025-03-18: **Anthropic Frontier Red Team** report shows Claude approaching undergraduate-level cyber skills[61]

February 2025

2025-02-23: **Claude 3.7 Sonnet** released as first hybrid reasoning model on market[62]

January 2025

2025-01-20: **DeepSeek R1** released with 671B params (37B active), reasoning optimizations[40]

2025-01-09: **Phi-4, Qwen2 VL 72B, Aya Expanse 32B** benchmarked on MMLU-Pro CS[63]

2025-01-01: **DeepSeek-V3** initially released, 671B params (37B active MoE)[39] [40]

December 2024

2024-12-26: **DeepSeek-V3** public release begins Chinese open-source surge[40]

H) IMPLEMENTATION EXAMPLES

1. GPT-5 with Adaptive Thinking (Python)

```
from openai import OpenAI
client = OpenAI(api_key="sk-...")

# Auto-routing between fast/deep reasoning
response = client.chat.completions.create(
    model="gpt-5",
    messages=[
        {"role": "system", "content": "You are a senior software architect."},
        {"role": "user", "content": "Design a distributed caching system for 1M QPS with <10ms
    ],
    verbosity="high",  # Controls response detail
    max_tokens=4000
)

# Force extended reasoning for hard problems
```

```
response_thinking = client.chat.completions.create(
    model="gpt-5-thinking",
    messages=[...],
    max_thinking_tokens=16000  # Budget for reasoning
)
```

2. Claude 4 Opus Long-Horizon Coding (Python)

```
import anthropic
client = anthropic.Anthropic(api_key="sk-ant-...")

# Enable extended thinking + tool use
response = client.messages.create(
    model="claude-opus-4-20250804",
    max_tokens=8192,
    thinking_budget="high",  # Allocate more reasoning compute
    tools=[
        {
            "name": "read_file",
            "description": "Read contents of file",
            "input_schema": {
                "type": "object",
                "properties": {"path": {"type": "string"}},
                "required": ["path"]
            }
        },
        {
            "name": "write_file",
            "description": "Write/edit file",
            "input_schema": {
                "type": "object",
                "properties": {
                    "path": {"type": "string"},
                    "content": {"type": "string"}
                },
                "required": ["path", "content"]
            }
        }
    ],
    messages=[
        {"role": "user", "content": "Refactor this 5000-line codebase to use async/await patte
    ]
)

# Opus 4 will create memory files for long sessions
```

3. Gemini 2.5 Pro Long-Context Analysis (Python)

```
import google.generativeai as genai
genai.configure(api_key="...")

model = genai.GenerativeModel(
    model_name="gemini-2.5-pro",
    generation_config={
        "max_output_tokens": 8192,
        "temperature": 0.7,
```

```
        "thinking_duration": "medium"  # Adaptive reasoning
    }
)

# Process entire codebase (2M tokens)
with open("full_repo.txt", "r") as f:
    codebase = f.read()  # 300K+ tokens

response = model.generate_content([
    "Analyze this entire codebase for security vulnerabilities, architectural issues, and tech
    codebase
])

# Gemini handles 2M context natively
```

4. DeepSeek-V3.2 Cost-Efficient Batch (Python)

```
import openai
client = openai.OpenAI(
    api_key="sk-...",
    base_url="https://api.deepseek.com/v1"
)

# Hybrid mode: force thinking with <think> tags
response = client.chat.completions.create(
    model="deepseek-v3-2-exp",
    messages=[
        {"role": "system", "content": "Use <think> tags for complex reasoning."},
        {"role": "user", "content": "Solve this dynamic programming problem: [...]"}
    ],
    temperature=0.0
)

# At $0.07/M (cache hit), 10× cheaper than GPT-5
```

5. Qwen3 Agent with MCP Tools (Python)

```
from qwen_agent import Agent, MCPServer

# Initialize MCP server for tool orchestration
mcp = MCPServer([
    {"name": "search", "endpoint": "https://api.tavily.com/search"},
    {"name": "calculator", "function": lambda x: eval(x)},
    {"name": "file_ops", "endpoint": "file:///workspace"}
])

# Qwen3 with native agent capabilities
agent = Agent(
    model="qwen3-235b-a22b-thinking-2507",
    mcp_server=mcp,
    thinking_mode="auto"  # Switches between thinking/non-thinking
)

result = agent.run(
    "Research top 5 competitors, calculate market share, generate comparison table in CSV"
)
```

```
# Qwen3 excels at multi-agent orchestration
```

6. Kimi K2 Agentic Coding (cURL API)

```
curl https://api.moonshot.cn/v1/chat/completions \
  -H "Authorization: Bearer sk-..." \
  -H "Content-Type: application/json" \
  -d '{
    "model": "kimi-k2-instruct",
    "messages": [
      {"role": "system", "content": "You are an expert full-stack developer."},
      {"role": "user", "content": "Build a React dashboard with real-time WebSocket data, dark
    ],
    "functions": [
      {
        "name": "execute_code",
        "description": "Run code in sandbox",
        "parameters": {"type": "object", "properties": {"code": {"type": "string"}}}
      }
    ],
    "temperature": 0.7,
    "max_tokens": 8000
  }'
```

7. Grok 4 Heavy Multi-Agent Research (Conceptual)

```
# Grok 4 Heavy uses multi-agent system internally
import xai
client = xai.Client(api_key="xai-...")

response = client.chat.completions.create(
    model="grok-4-heavy",  # Multi-agent mode
    messages=[
        {"role": "user", "content": "Solve this PhD-level quantum mechanics problem with full
    ],
    max_thinking_time=300,  # 5 minutes reasoning budget
    temperature=0.0
)

# Heavy mode coordinates multiple reasoning agents
# Achieves 50.7% on Humanity's Last Exam
```

8. Multi-Model Router (LangChain + LiteLLM)

```
from litellm import completion
import anthropic
import openai

def route_query(query: str, context_length: int, budget: str):
    """Smart routing based on task requirements"""

    # Quick answers → Gemini Flash
    if len(query) < 100 and budget == "low":
```

```
        return completion(
            model="gemini/gemini-2.5-flash",
            messages=[{"role": "user", "content": query}]
        )

    # Long context → Gemini Pro
    elif context_length > 128000:
        return completion(
            model="gemini/gemini-2.5-pro",
            messages=[{"role": "user", "content": query}],
            max_tokens=8192
        )

    # Sustained coding → Claude Opus
    elif "refactor" in query.lower() or "debug" in query.lower():
        return completion(
            model="anthropic/claude-opus-4-20250804",
            messages=[{"role": "user", "content": query}],
            thinking_budget="high"
        )

    # Budget batch → DeepSeek
    elif budget == "low":
        return completion(
            model="deepseek/deepseek-v3-2-exp",
            messages=[{"role": "user", "content": query}]
        )

    # Default → GPT-5
    else:
        return completion(
            model="openai/gpt-5",
            messages=[{"role": "user", "content": query}],
            verbosity="medium"
        )

# Usage
result = route_query(
    query="Explain quantum entanglement in 50 words",
    context_length=150,
    budget="low"
)
```

DETAILED MODEL PROFILES

1. GPT-5 (OpenAI) — Universal Frontier Leader

Release: August 7, 2025[3] [1]

Architecture: Unified system with `gpt-5-main` (fast), `gpt-5-thinking` (reasoning), automatic router[29] [30]

Key Benchmarks:

- **AIME 2025**: 94.6% (no tools), 88.9% (o3), industry-leading[1] [2]
- **SWE-bench Verified**: 74.9% (vs 69.1% o3, 54.6% GPT-4.1)[64] [2]

- **GPQA Diamond**: 87.3% with tools, 85.7% without[1] [2]
- **MMMU**: 84.2% (multimodal understanding)[42] [1]
- **HealthBench Hard**: 46.2% (up from 31.6% o3)[65] [1]
- **FrontierMath**: 26.3% (expert-level math with Python)[2]

Strengths:

- **Automatic thinking mode** selection eliminates user complexity[30] [29] [3]
- **Verbosity control** (`low/medium/high`) for response length tuning[49] [2]
- **Parallel tool calling**: 40+ simultaneous tool calls, 96.7% τ2-bench telecom[1] [2]
- **26% lower hallucination** rate vs GPT-4o[66] [59]
- **Multimodal excellence**: Text, image, video, audio, charts[42] [3] [1]

Weaknesses:

- **Highest cost**: $1.25 input, $10 output per M tokens[42]
- **Latency**: 9.98 sec P50 (main), slower than GPT-4o mini[42]
- **Closed-source**: No weights, limited customization[3]

Best For: Organizations needing best-in-class accuracy across diverse tasks (legal, health, research, coding) with budget flexibility[56] [66] [1]

Applications: Expert-level work (50% of cases), complex reasoning, multi-step agentic workflows, enterprise knowledge work[56] [1]

2. Claude 4 Opus (Anthropic) — Long-Horizon Coding Champion

Release: May 22, 2025[6] [5]

Architecture: Opus 4.1 upgrade (Aug 4, 2025) with extended thinking + tool use[4]

Key Benchmarks:

- **SWE-bench Verified**: 74.5% (Opus 4.1), state-of-the-art real-world coding[4] [5]
- **Terminal-bench**: 43.2% (command-line tasks)[5]
- **GPQA Diamond**: Competitive with GPT-5 on PhD-level science[5]
- **Multi-file refactoring**: GitHub reports "notable gains" over Opus 4[67] [4]

Strengths:

- **Sustained focus**: 30+ hours continuous performance without degradation[45] [25] [5]
- **Precision debugging**: Pinpoint exact corrections in large codebases without collateral changes[67] [4]
- **Memory capabilities**: Autonomously creates "memory files" for long-term task awareness[5]
- **Safety alignment**: Perfect 1.000 on Password Protection, superior to competitors[57]
- **Clean code generation**: Self-documenting, well-factored output, especially for C++[68] [5]

Weaknesses:

- **Highest API cost**: $15 input, $75 output (Opus 4)[4]
- **Slower than Sonnet**: Not ideal for latency-sensitive applications[5]
- **Less multilingual**: Strong English, but fewer languages than Gemini/Qwen[35]

Best For: Professional developers, large-scale refactors, enterprise coding agents, safety-critical applications[45] [4] [5]

Applications: GitHub Copilot, Cursor IDE agents, Replit code generation, multi-hour autonomous coding sessions[45] [5]

3. Gemini 2.5 Pro (Google DeepMind) — Long-Context Multimodal King

Release: March 25, 2025 (GA: June 16, 2025)[7] [9] [50]

Architecture: Hybrid reasoning model with thinking budgets, native multimodality (early fusion)[9] [7] [8]

Key Benchmarks:

- **Context**: 2M tokens production (testing 6 hours video)[7] [8]
- **VideoMME**: 84.8% (best video understanding)[47] [7]
- **WebDev Arena**: +147 Elo over previous version (UI development)[7]
- **SWE-bench Verified**: 63.8% (with custom agent setup)[69]
- **AIME 2025**: 86.7% (math reasoning)[69]
- **GPQA Diamond**: 84% (science)[69]

Strengths:

- **Unmatched context**: Only production-ready 2M token model[32] [8] [7]
- **Native multimodality**: Text, image, audio, video seamlessly integrated[47] [9] [7]
- **Spatial understanding**: Best for robotics, object detection from camera feeds[32]
- **Adaptive thinking**: Budgeted reasoning (low/medium/high duration)[9] [50]
- **Real-time grounding**: Google Search integration for up-to-date info[8]

Weaknesses:

- **Code execution limits**: Lags Claude/GPT-5 on pure coding benchmarks[69]
- **Pricing complexity**: Varies by tier, not always transparent[7]
- **Privacy concerns**: Type IV failure in red team testing (geolocation disclosure)[70]

Best For: Full repository analysis, legal document review, video content understanding, long-form research[32] [8] [7]

Applications: Gemini app with Canvas, video analysis, large-scale document intelligence, academic research[71] [9] [7]

4. DeepSeek-V3.1 / V3.2-Exp (DeepSeek) — Cost-Efficiency Leader

Release: V3.1 (Aug 29, 2025), V3.2-Exp (Sep 29, 2025)[12] [10] [11]

Architecture: 671B params (37B active MoE), sparse attention (DSA in V3.2), hybrid thinking/non-thinking[10] [11] [12]

Key Benchmarks:

- **AIME 2025**: 88.4% (thinking mode, edges out R1)[10]
- **LiveCodeBench**: 74.8% (ahead of V3)[10]
- **MMLU-Pro**: Competitive with GPT-4.1[10]
- **V3.2 performance**: On par with V3.1 despite efficiency gains[72] [73] [12]

Strengths:

- **Lowest API cost**: $0.07/M input (cache hit), $0.28 (miss), $0.42 output—50%+ cut from V3.1[11] [12]
- **Hybrid modes**: Seamless switching between fast/reasoning within single model[11] [10]
- **Open-source**: MIT license, full weights available[11] [10]
- **128K context**: Supports long conversations and code analysis[11] [10]
- **Sparse attention**: DSA reduces compute in long-context scenarios[73] [12] [11]

Weaknesses:

- **API stability**: Slowdowns reported during peak usage (pre-V3.1)[10]
- **Thinking verbosity**: Can over-reason simple queries, increasing latency[10]
- **Documentation**: Less mature than OpenAI/Anthropic ecosystems[10]

Best For: Budget-conscious startups, high-volume batch jobs, research teams, China-focused deployments[74] [11] [10]

Applications: Day-one support for Azure, RunPod, SGLang, vLLM; widely adopted for cost-sensitive workflows[75]

5. Qwen3-235B-A22B (Alibaba Cloud) — Multilingual Agent Champion

Release: April 28, 2025 (Qwen3), July 2025 (235B-Thinking)[13] [14] [51]

Architecture: 235B params (22B active MoE), hybrid reasoning, 119 languages[14] [13] [33]

Key Benchmarks:

- **MMLU-Pro**: 83.0% (knowledge)[33]
- **AIME25**: 70.3% (math reasoning)[33]
- **LiveCodeBench v6**: 51.8% (coding)[33]
- **BFCL-v3**: 70.9% (function calling/tool use)[33]
- **Arena-Hard v2**: 82.7% (human preference, GPT-4.1 judged)[33]

Strengths:

- **Best multilingual**: 119 languages with strong non-English instruction-following[15] [13] [14]
- **Agent capabilities**: Leading open-source for tool orchestration, MCP native support[51] [15] [33]
- **Hybrid reasoning**: Automatic switching between thinking/non-thinking modes[52] [13] [14]
- **Context window**: 262K tokens native, extensible to 1M+ with YaRN RoPE[52] [33]
- **Cost-efficient**: $0.27 input, $1.10 output per M tokens[43]

Weaknesses:

- **Smaller than frontier closed models**: Trails GPT-5, Claude on some benchmarks[33]
- **Less mature ecosystem**: Fewer third-party integrations than OpenAI[33]
- **Thinking mode**: Not always as deep as GPT-5 Pro or Claude thinking[33]

Best For: Global enterprises, multilingual customer service, agent-based workflows, research labs[13] [15] [33]

Applications: Customer support (23 languages), data analysis agents, academic research, SEA/MENA markets[15] [52]

6. Kimi K2 (Moonshot AI) — Agentic Coding Disruptor

Release: July 2025[17] [16] [18]

Architecture: 1T params (32B active MoE), MuonClip optimizer, 384 experts, 128K context[76] [16] [17]

Key Benchmarks:

- **LiveCodeBench v6**: 53.7% (beats GPT-4.1's 44.7%)[16] [17]
- **SWE-bench Verified**: 65.8% single-attempt, 71.6% with retries[17]
- **MATH-500**: 97.4% (math problems)[17]
- **Agentic demos**: 16-step salary analysis, 17-step concert planning with tool use[17]

Strengths:

- **Ultra-low cost**: $0.15 input, $2.50 output—undercutting OpenAI/Anthropic 10×[38] [17]
- **Agentic optimization**: Specifically trained for tool use, function calling, multi-step workflows[38] [16] [17]
- **MuonClip training**: 2× training efficiency, 50% less memory, zero failures over 15.5T tokens[17]
- **Open-source**: Fully available on HuggingFace[16] [17]

Weaknesses:

- **Lukewarm adoption**: <200K downloads vs 400K+ for DeepSeek-V3 (as of July 2025)[16]
- **Prompt engineering**: Requires careful iteration for optimal results[18]
- **Latency**: Slower than advertised for 1T param model, recommend streaming[18]
- **Documentation**: Less clear than established providers[18]

Best For: Experienced developers, agentic coding workflows, budget-constrained teams, researchers[16] [18] [17]

Applications: Code generation (Deepsite, AnyCoder), web dev, autonomous software agents[18] [16]

7. Llama 4 Maverick & Scout (Meta AI) — Open Multimodal Powerhouse

Release: April 5, 2025[19] [20] [37]

Architecture: First Llama MoE (Maverick: 400B total/17B active, 128 experts; Scout: 109B/17B, 16 experts)[20] [19]

Key Benchmarks:

- **Context**: 10M tokens (Scout), 1M (Maverick)—largest in Llama family[21] [19] [20]
- **Languages**: 200 languages trained, 12 fine-tuned (Arabic, Spanish, German, Hindi, etc.)[37] [19]
- **Multimodal**: Native text + image input via early fusion[19] [37]

Strengths:

- **Open-weight**: Free under 700M MAU, full access to weights[44] [21] [19]
- **Massive context**: Scout's 10M tokens enables full book/codebase analysis[20] [19]
- **Native multimodality**: No separate encoders, unified understanding[37] [19]
- **Meta support**: Proven infrastructure, Azure, AWS, GCP integrations[77] [36]
- **Efficient deployment**: Scout fits single H100 via quantization[19]

Weaknesses:

- **License restriction**: Requires separate license above 700M MAU[21] [44]
- **Scout experimental**: 10M context still being tested for reliability[20] [19]
- **Benchmarks lag**: Behind GPT-5, Claude on most leaderboards[78] [19]
- **Image understanding**: English-only for vision tasks (as of April 2025)[21]

Best For: Startups, researchers, open-source community, enterprises below MAU threshold[44] [21] [19]

Applications: Azure AI Foundry, Databricks, on-prem deployments, distillation base[36] [78]

8. Grok 4 (xAI) — PhD-Level Reasoning Specialist

Release: July 9, 2025[22] [48] [23]

Architecture: Axiom-based first-principles reasoning, Colossus supercomputer training, 256K context[22] [23]

Key Benchmarks:

- **Humanity's Last Exam**: 50.7% (Heavy mode, multi-agent)—first to cross 50%[48] [22]
- **HMMT 2025**: 100% (Harvard-MIT math tournament)[24] [22]
- **ARC-AGI v1**: 66.6% (abstract reasoning)[24] [22]
- **Vending-Bench**: $3,331 net worth, 2,271 units sold (2× nearest competitor)[22]
- **GPQA Diamond**: 90.0% (PhD-level science)[24] [22]

Strengths:

- **Test-time compute scaling**: Heavy mode uses multi-agent coordination for 2-3× better performance[23] [22]
- **Academic focus**: Best on hardest reasoning benchmarks (HLE, ARC-AGI)[48] [22] [24]
- **Business simulation**: Excels at long-horizon planning (Vending-Bench)[22]
- **Real-time X integration**: Access to live social data[48] [23]

Weaknesses:

- **Slow**: Heavy mode 3-5× slower than standard, not for production apps[23] [22]
- **Expensive**: $3 input, $15 output; Heavy requires special access[23]
- **Limited context**: 256K (vs 2M Gemini, 10M Scout)[22] [23]
- **Early multimodal**: Vision/image gen "coming soon" as of July 2025[23]

Best For: Academic research, PhD-level problem-solving, long-term strategic planning[24] [22] [23]

Applications: Scientific research, complex simulations, strategic business modeling[48] [22]

9. Claude Sonnet 4.5 (Anthropic) — Balanced Efficiency Leader

Release: September 28, 2025[25]

Architecture: Sonnet 4 (May 22, 2025) → Sonnet 4.5 (Sep 28, 2025), optimized for efficiency[25] [5]

Key Benchmarks:

- **SWE-bench Verified**: 72.7% (Sonnet 4)[5]
- **OSWorld**: 61.4% (computer use, up from 42.2% 4 months prior)[25]
- **Sustained performance**: 30+ hours continuous focus maintained[25] [5]

Strengths:

- **Best computer use**: State-of-the-art browser navigation, form filling, spreadsheet automation[25]
- **Cost-performance balance**: Lower than Opus, better than most alternatives[25] [5]
- **Faster than Opus**: Optimal for high-volume production use[5]
- **Coding excellence**: GitHub reports "state-of-the-art" for agentic coding[5]

Weaknesses:

- **Not quite Opus**: Slightly lower raw intelligence in most domains[79] [5]
- **Less documented**: Sonnet 4.5 benchmarks not fully published yet[25]
- **Still expensive**: Higher than open-source options like DeepSeek/Qwen[25]

Best For: Production coding agents, computer use automation, balanced cost-performance needs[25] [5]

Applications: GitHub Copilot, Cursor, Sourcegraph Cody, iGent autonomous dev[5]

10. Mistral Medium 3 (Mistral AI) — Enterprise Cost-Optimizer

Release: May 7, 2025[28] [26]

Architecture: Optimized dense model, 4-GPU minimum deployment[28] [26]

Key Benchmarks:

- **Performance**: 90% of Claude Sonnet 3.7 across benchmarks[26]
- **Cost**: $0.4 input, $2 output per M tokens—8× cheaper than comparable models[26]
- **Deployment**: 4× H100 minimum, VPC/on-prem support[28] [26]

Strengths:

- **Best cost-performance**: 8× cheaper while maintaining near-frontier quality[26]
- **On-prem deployment**: Hybrid cloud, in-VPC, custom post-training options[28] [26]
- **Enterprise features**: Integration with enterprise tools, custom training[26]
- **Fast inference**: Optimized for production latency requirements[26]

Weaknesses:

- **Not quite frontier**: Trails GPT-5, Claude Opus on hardest benchmarks[26]
- **Smaller ecosystem**: Fewer integrations than OpenAI/Anthropic[26]
- **GPU requirements**: 4-GPU minimum limits accessibility[26]

Best For: Enterprises needing on-prem deployment, cost-conscious teams, European data sovereignty[28] [26]

Applications: Enterprise document processing, customer service, internal tools, regulated industries[28] [26]

KEY TAKEAWAYS & STRATEGIC RECOMMENDATIONS

For Enterprises:

1. **Multi-model strategy is table stakes** — No single model dominates all use cases[80] [81] [82]
2. **GPT-5 + Claude Opus** covers 80% of needs (general + sustained coding)[1] [4] [5]
3. **Add Gemini 2.5 Pro** for long-context (legal, video, research)[8] [7]
4. **DeepSeek-V3.2** for cost-sensitive internal tools (50%+ savings)[12] [11]
5. **Mistral Medium 3** for on-prem/regulated environments[28] [26]

For Startups:

1. **Start with Qwen3 or DeepSeek** — Open-source, cost-efficient, capable[13] [11] [33] [10]
2. **Add Claude Sonnet** for production coding (balanced cost/performance)[25] [5]
3. **Use Gemini Flash** for latency-sensitive queries (<5 sec)[9]
4. **Kimi K2** for agentic experiments ($0.15/$2.50)[16] [17]

For Researchers:

1. **Grok 4 Heavy** for hardest reasoning problems[24] [48] [22]
2. **GPT-5 Pro** for math/science with extended thinking[2] [1]
3. **Qwen3 or Llama 4** for open experimentation[13] [20] [33]
4. **Gemini 2.5 Pro** for multimodal/video analysis[47] [9] [7]

Architecture Evolution (2026+ Outlook):

- **Hybrid reasoning** becomes standard (thinking/non-thinking in single model)[29] [13] [10]
- **Context windows** push toward 10M+ tokens (Llama 4 Scout leading)[19] [20]
- **MoE efficiency** enables 1T+ param models at <100B active compute[31] [17]
- **Agentic optimization** shifts from general intelligence to specialized tool use[17] [16] [33]
- **Multimodal native** replaces separate encoders universally[37] [7] [19]

Report compiled: Friday, October 24, 2025, 8:25 AM AWST
Sources: 151 citations from official docs, release notes, benchmarks, and technical reports[1-151]
Next refresh: November 2025 (monthly cadence recommended for fast-moving LLM landscape)

⁂

1. https://openai.com/index/introducing-gpt-5/
2. https://openai.com/index/introducing-gpt-5-for-developers/
3. https://openai.com/gpt-5/
4. https://www.anthropic.com/news/claude-opus-4-1
5. https://www.anthropic.com/news/claude-4
6. https://www.datacamp.com/blog/claude-4
7. https://blog.google/products/gemini/gemini-2-5-pro-updates/
8. https://huggingface.co/blog/proflead/google-gemini-2-5-pro
9. https://blog.google/products/gemini/gemini-2-5-model-family-expands/
10. https://felloai.com/2025/08/deepseek-v3-1-is-here-chinese-most-advanced-open-source-ai-yet/
11. https://www.bentoml.com/blog/the-complete-guide-to-deepseek-models-from-v3-to-r1-and-beyond
12. https://api-docs.deepseek.com/news/news250929
13. https://huggingface.co/Qwen/Qwen3-32B
14. https://huggingface.co/Qwen/Qwen3-0.6B
15. https://www.deploy.ai/blog-post/qwen-3-by-alibaba-cloud-everything-you-need-to-know
16. https://huggingface.co/blog/freddyaboulton/no-bs-kimmik2
17. https://huggingface.co/blog/fdaudens/moonshot-ai-kimi-k2-explained
18. https://www.datacamp.com/tutorial/kimi-k2
19. https://huggingface.co/blog/llama4-release
20. https://www.deploy.ai/blog-post/llama-4-by-meta-ai-everything-you-need-to-know
21. https://www.techtarget.com/whatis/feature/Meta-Llama-4-explained-Everything-you-need-to-know
22. https://www.datacamp.com/blog/grok-4

23. https://yourgpt.ai/blog/updates/grok-4
24. https://www.deeplearning.ai/the-batch/grok-4-launches-with-benchmark-records-and-idiosyncratic-behavior/
25. https://www.anthropic.com/news/claude-sonnet-4-5
26. https://mistral.ai/news/mistral-medium-3
27. https://www.promptingguide.ai/models/mistral-large
28. https://mistral.ai/models
29. https://academy.openai.com/public/clubs/work-users-ynjqu/resources/models
30. https://academy.openai.com/public/clubs/work-users-ynjqu/resources/intro-gpt-5
31. https://huggingface.co/blog/Kseniase/moe2
32. https://kartaca.com/en/unlocking-new-horizons-a-deep-dive-into-gemini-2-5s-multimodal-capabilities/
33. https://huggingface.co/Qwen/Qwen3-Next-80B-A3B-Instruct
34. https://huggingface.co/Qwen/Qwen3-Next-80B-A3B-Instruct-FP8
35. https://www.anthropic.com/news/claude-3-family
36. https://azure.microsoft.com/en-us/blog/introducing-the-llama-4-herd-in-azure-ai-foundry-and-azure-databricks/
37. https://ai.meta.com/blog/llama-4-multimodal-intelligence/
38. https://www.thoughtworks.com/en-au/insights/blog/generative-ai/kimi-k2-whats-fuss-whats-like-use
39. https://huggingface.co/blog/wolfram/llm-comparison-test-2025-01-02
40. https://explodingtopics.com/blog/list-of-llms
41. https://backlinko.com/list-of-llms
42. https://www.leanware.co/insights/gpt-5-features-guide
43. https://www.sobot.io/blog/compare-llm-models-2025/
44. https://zapier.com/blog/llama-meta/
45. https://www.reddit.com/r/ClaudeAI/comments/1ksz4r0/claude_4_opus_is_actually_insane_for_coding/
46. https://www.emergentmind.com/topics/gemini-2-5-pro
47. https://huggingface.co/blog/FOUND-AI/found-symbiotic-architecture-for-narrative-video
48. https://x.ai/news/grok-4
49. https://cdn.openai.com/pdf/47c0215b-8976-4f60-8e13-d69c2ddbc15e/a-practical-guide-to-building-with-gpt-5.pdf
50. https://blog.google/technology/google-deepmind/gemini-model-thinking-updates-march-2025/
51. https://qwenlm.github.io/blog/qwen3/
52. https://momen.app/blogs/qwen3-main-features-hybrid-reasoning-multilingual-support/
53. https://www.anthropic.com/news/salesforce-anthropic-expanded-partnership
54. https://reflection.ai/blog/frontier-open-intelligence
55. https://huggingface.co/blog/bigcode/arena
56. https://openai.com/index/gdpval/
57. https://openai.com/index/openai-anthropic-safety-evaluation/
58. https://epoch.ai/data-insights/consumer-gpu-model-gap
59. https://cdn.openai.com/gpt-5-system-card.pdf
60. https://openai.com/index/gpt-5-system-card/

61. https://www.anthropic.com/news/strategic-warning-for-ai-risk-progress-and-insights-from-our-frontier-red-team
62. https://www.anthropic.com/news/claude-3-7-sonnet
63. https://huggingface.co/blog/wolfram/llm-comparison-test-2025-01-10
64. https://www.datacamp.com/blog/gpt-5
65. https://www.datastudios.org/post/chatgpt-5-full-report-on-models-features-capabilities-performance-pricing-and-more
66. https://www.solutelabs.com/blog/gpt-5-guide
67. https://blog.getbind.co/2025/08/06/claude-opus-4-1-vs-claude-opus-4-how-good-is-this-upgrade/
68. https://community.openai.com/t/i-cancelled-my-plus-subscription-and-im-switching-to-claude-sonnet-3-7-for-c-coding/1246558
69. https://www.techtarget.com/whatis/feature/Google-Gemini-25-Pro-explained-Everything-you-need-to-know
70. https://huggingface.co/blog/unmodeled-tyler/persona-collapse-in-llms
71. https://blog.google/products/gemini/gemini-collaboration-features/
72. https://skywork.ai/blog/deepseek-3-2-vs-3-1-terminus-comparison-2025/
73. https://dev.to/czmilo/deepseek-v32-exp-complete-analysis-2025-ai-model-breakthrough-and-in-depth-analysis-of-sparse-3gcl
74. https://www.nist.gov/system/files/documents/2025/09/30/CAISI_Evaluation_of_DeepSeek_AI_Models.pdf
75. https://huggingface.co/blog/paresh2806/sglang-efficient-llm-workflows
76. https://github.com/MoonshotAI/Kimi-K2
77. https://huggingface.co/blog/dell-ai-applications
78. https://www.datacamp.com/blog/llama-4
79. https://www.edenai.co/post/claude-sonnet-4-vs-claude-opus-4
80. https://alphacorp.ai/best-llms-for-october-2025/
81. https://codedesign.ai/blog/the-ultimate-guide-to-the-top-large-language-models-in-2025/
82. https://inkeep.com/blog/best-ai-models-2025
83. https://towardsdatascience.com/how-to-use-frontier-vision-llms-qwen-3-vl-2/
84. https://www.shakudo.io/blog/top-9-large-language-models
85. https://artificialanalysis.ai/leaderboards/models
86. https://www.vellum.ai/llm-leaderboard
87. https://botpress.com/blog/best-large-language-models
88. https://www.jwatch.org/na58969/2025/08/14/large-language-models-and-clinical-reasoning-frontier-2025
89. https://felloai.com/2025/10/the-best-ai-in-october-2025-we-compared-chatgpt-claude-grok-gemini-others/
90. https://www.stephensonharwood.com/insights/neural-network-october-2025
91. https://blog.google/products/gemini/google-ai-pro-students-learning/
92. https://www.anthropic.com/news/claude-3-5-sonnet
93. https://www.anthropic.com/news/anthropic-education-report-how-university-students-use-claude
94. https://deepmind.google/models/gemini/pro/
95. https://huggingface.co/blog/vlms-2025
96. https://openai.com/index/estimating-worst-case-frontier-risks-of-open-weight-llms/
97. https://huggingface.co/blog/driaforall/mem-agent-blog

98. https://openai.com/global-affairs/our-approach-to-frontier-risk/
99. https://openai.com/index/better-language-models/
100. https://cdn.openai.com/threat-intelligence-reports/7d662b68-952f-4dfd-a2f2-fe55b041cc4a/disrupting-malicious-uses-of-ai-october-2025.pdf
101. https://huggingface.co/blog/dvilasuero/choosing-best-open-source-ai-models
102. https://openai.com/index/ai-clinical-copilot-penda-health/
103. https://community.openai.com/t/architectural-flaws-in-modern-llm-systems-we-need-to-talk/1362796
104. https://huggingface.co/blog/Tomedes/best-ai-assitants-and-agents
105. https://huggingface.co/blog/isaacus/introducing-mleb
106. https://openai.com/index/defining-and-evaluating-political-bias-in-llms/
107. https://huggingface.co/blog/math_verify_leaderboard
108. https://www.splunk.com/en_us/blog/learn/llms-best-to-use.html
109. https://arxiv.org/abs/2510.07024
110. https://www.reddit.com/r/LocalLLaMA/comments/1obqkpe/best_local_llms_october_2025/
111. https://www.instaclustr.com/education/open-source-ai/top-10-open-source-llms-for-2025/